# STUDENT STEP BY STEP PROCEDURE MANUAL

### Fundamentals of Nursing & Clinical Skills Lab (1)

## SKILL: Hand Washing (Medical Asepsis)

| Student Name | Student ID | Instructor | Date |
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## 1. Purpose

To remove transient microorganisms, reduce the total number of resident microbial flora on the hands, prevent the transmission of healthcare associated infections (HAIs), and maintain a safe, aseptic clinical environment.

## 2. Key Principles

- **Friction & Mechanical Action:** Friction is the single most effective element in removing dirt, oils, and transient bacteria from skin folds and surfaces.
- **Direction of Flow:** Water should flow from clean areas (wrists/forearms) toward the dirtiest area (fingertips) during surgical scrubs, but for routine medical asepsis, hands are kept downward so water flows from fingertips down.
- **Infection Control Barrier:** Hand hygiene is universally recognized as the single most critical practice to reduce cross-contamination and protect both clients and healthcare providers.

## 3. Required Equipment / Supplies

- Accessible sink with warm running water (temperature controlled by foot, knee, or motion sensors preferred).
- Antimicrobial liquid soap or plain soap in a disposable dispenser.
- Clean paper towels for drying hands.
- Waste receptacle for paper towel disposal.
- Hand lotion or cream (optional, to prevent skin dryness/chapping).

## 4. Step by Step Procedure

| Step | Detailed Procedure | Clinical Rationale & Guidance | Practice Check |
|---|---|---|---|
| **1** | Stand in front of the sink, ensuring your clothing and lab coat do not touch the sink basin or surrounding surfaces. | Prevents contamination of clean uniform from micro-organisms on the sink rim. | ☐ |
| **2** | Remove all rings, watches, and bracelets from hands and wrists. (Short, natural nails with no artificial nails or polish). | Jewelry harbors microorganisms and prevents thorough cleansing of skin crevices; long nails tear gloves. | ☐ |

| Step | Detailed Procedure | Clinical Rationale & Guidance | Practice Check |
|---|---|---|---|
| | | | |
| **3** | Turn on water faucet and adjust flow to achieve a comfortable, warm (not hot) water temperature. | Warm water opens skin pores and removes natural oils less harshly than hot water. | ☐ |
| **4** | Wet hands and wrists thoroughly under running water, keeping hands lower than elbows at all times. | Ensures water flows downward from cleanest area (wrists) to fingertips, preventing contamination of arms. | ☐ |
| **5** | Apply 3 to 5 mL of liquid antimicrobial soap to your hands, covering all skin surfaces. | Sufficient soap is required to lather and bind with oils and transient microbes. | ☐ |
| **6** | Rub hands vigorously together, palm to palm, covering all surfaces with a rich lather. | Initiates mechanical friction to loosen dirt and transient microorganisms. | ☐ |
| **7** | Interlace fingers and rub palms back and forth with fingers interlocked. | Cleanses interdigital spaces thoroughly where bacteria frequently accumulate. | ☐ |
| **8** | Place right palm over left dorsum with interlaced fingers, then switch hands and repeat. | Ensures complete friction and cleansing of the back of both hands. | ☐ |
| **9** | Clasp each thumb in the opposite palm using a rotational rubbing motion, and rub fingertips in circular motions against the opposite palm. | Cleanses high-risk areas including thumbs, nail beds, and fingertips. | ☐ |
| **10** | Continue friction scrubbing for a minimum of **20 seconds** across all hand surfaces. | A minimum of 20 seconds of mechanical friction is required to effectively reduce microbial load. | ☐ |
| **11** | Rinse hands thoroughly under running water, keeping fingers pointing downward. | Removes loosened soap, dirt, and microorganisms completely from skin surfaces without contaminating arms. | ☐ |
| **12** | Dry hands thoroughly with a clean paper towel, starting from fingertips and moving upward toward the wrists and forearms. | Drying from clean fingers to forearms prevents skin irritation and re-contamination. | ☐ |
| **13** | Use a fresh, dry paper towel to turn off the water faucet, avoiding direct hand contact with handles. | Prevents recontaminating clean hands by touching unwashed, contaminated faucet handles. | ☐ |
| **14** | Discard used paper towels immediately into the designated waste receptacle. | Maintains environmental hygiene and proper biohazard/waste control. | ☐ |
| **15** | Inspect hands to ensure cleanliness and apply hand lotion if necessary to maintain skin integrity. | Maintains skin barrier health, preventing cracks and dermatitis caused by frequent washing. | ☐ |

## 5. Remember

| 20 Seconds Minimum | Fingers Down & Paper Towel Faucet |
|---|---|
| Always scrub hands vigorously with soap for at least **20 seconds** to ensure effective microbial removal. | Keep fingers pointing downward during rinsing and use a paper towel to turn off the water faucet. |

## 6. Critical Safety Points

- Apply Hand Washing according to WHO's Five Moments for Hand Hygiene which are: 1- Before touching a patient, 2- Before a clean/aseptic procedure, 3- After body-fluid exposure or risk, 4- After touching a patient, and 5- After touching the patient's surroundings.
- Never touch the inside of the sink basin or allow your uniform to touch the sink during hand washing.
- Always remove rings, watches, and bracelets before beginning hand hygiene.
- Never skip cleaning inter digital spaces, thumbs, and fingernail beds.
- Always use a clean paper towel to turn off manual faucets to prevent recontamination.
- Perform hand hygiene before and after every client contact, procedure, or removal of gloves.

## 7. Common Errors to Avoid

- Failure to mention the WHO's Five Moments for Hand Hygiene which are: 1- Before touching a patient, 2- Before a clean/aseptic procedure, 3- After body-fluid exposure or risk, 4- After touching a patient, and 5- After touching the patient's surroundings.
- Washing hands for less than the recommended 20 seconds.
- Failing to remove rings and watches, which harbor high concentrations of bacteria.
- Holding hands upward during rinsing, allowing contaminated water to run back down toward the arms.
- Touching the water faucet handles with clean hands after washing.
- Using shared, cloth hand towels instead of single use disposable paper towels.
- Failing to dry hands completely, which promotes microbial transfer on moist skin surfaces.

## 8. Student Self Reflection Notes:

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